PRINT AND LEARN · ABOUT 20 MINUTES WITH PENCIL

Math Primer

Name: _____ Date: _____

Eleven ideas from everyday maths, in plain words, each with a tiny example you can check by hand. No prior maths needed beyond multiplying and adding.

THE THREAD THAT CONNECTS THEM

Numbers describe things, comparisons score them, tables transform them, and small repeated adjustments find the settings that work.

How the ideas fit together

DESCRIBE
vectors, and the multidimensional space they live in

COMPARE
distance, angle, dot product

TRANSFORM
matrix multiplication

TURN INTO SHARES
softmax, weighted choice

SHAPE AND STEADY
bends, rescaling

IMPROVE
error and nudging

REPEAT
feedback loops

READ TOP TO BOTTOM: EACH IDEA USES THE ONES BEFORE IT

Eleven ideas

Each card has the same beats: what it is, a small worked example, and an everyday automation comparison.

1 Vector

A **vector** is a list of numbers that describes one thing. The order is fixed: position 1 always means the same thing, so two vectors can be compared position by position.

```
ticket = [urgency 7, order value 3, age 5]
```

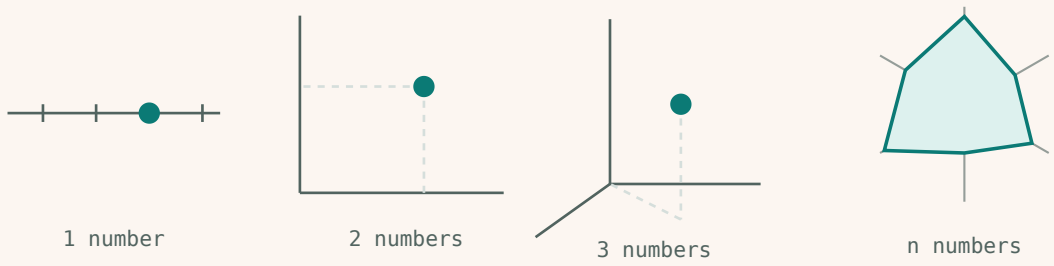
THINK OF IT AS A spreadsheet row where every column is a number.

YOUR TURN

Write a vector for a parcel: weight 4 kg, distance 120 km, 2 days old.

2 Multidimensional space

Every number in a vector is a coordinate along its own **axis**. One number places a point on a line, two on a flat plane, three in a room. A vector with **n** numbers places a point in **n-dimensional space**. You can only draw two or three axes, but positions, distances and angles follow the same rules with 10 axes or 1,000.



Why use so many axes? Each extra axis is one more way for two things to differ. Two cases can look identical on two measures and still be far apart on a third, so more axes allow finer distinctions.

```
[3, 4] and [3, 4] are the same point
```

```
[3, 4, 0] and [3, 4, 9] are 9 apart once a third axis is added
```

THINK OF IT AS

A spreadsheet with n numeric columns: every row is a point, and every column is an axis.

YOUR TURN

A point has 5 numbers. How many axes does its space have?

3 Closeness: distance and angle

To compare two points, measure the **distance** between them: in any number of dimensions, subtract position by position, square each difference, add them up and take the square root. Or compare the **angle** between the arrows from the origin: do they point the same way?

`[2, 5, 1] to [2, 6, 3]: differences [0, 1, 2] → √(0 + 1 + 4) ≈ 2.2`

`[2, 5, 1] to [9, 1, 8]: differences [7, −4, 7] → √114 ≈ 10.7`

`[1, 1] and [2, 2] point the same way; [1, 0] and [0, 1] are at right angles`

THINK OF IT AS

Plot every past case as a point. A new case that lands near a cluster probably behaves like that cluster.

YOUR TURN

Which is closer to [0, 0]: [3, 4] or [6, 1]?

4 Dot product

Multiply two vectors position by position, then add up. You get **one number** that says how well they match. Big and positive means they point the same way, near zero means unrelated, negative means opposite.

```
[2, 1, 0] · [1, 3, 5] = 2×1 + 1×3 + 0×5 = 5
```

THINK OF IT AS

A lead score: each fact about a lead is multiplied by how much it matters, then added up.

YOUR TURN

Work out [3, 0, 2] · [1, 5, 4].

5 Matrix multiplication

A **matrix** is a table of numbers. Multiplying a vector by a matrix takes the dot product with each column and gives a new vector, possibly of a different length. The shapes must line up: (1 × 3) times (3 × 2) gives (1 × 2).

```
[1, 2, 3] × [[1, 0], [0, 1], [1, 1]] = [1+0+3, 0+2+3] = [4, 5]
```

THINK OF IT AS

One mapping table applied to a whole row at once. Each output column is a scoring rule, and the matrix holds all the rules together.

YOUR TURN

Work out [2, 1] × [[1, 3], [4, 0]].

6 Softmax

Softmax turns any list of scores into percentages that add up to 100%. The highest score gets the biggest share, and nobody gets exactly 0% or 100%. Dividing the scores by a number above 1 first flattens the shares, and below 1 sharpens them.

```
scores [2, 1, 0] → 67% | 24% | 9%
```

THINK OF IT AS

Several classifiers each give a confidence. Softmax turns them into one set of shares you can route on.

YOUR TURN

Three equal scores [0, 0, 0] go through softmax. What are the shares?

7 Probability and weighted choice

If the shares are 50%, 30% and 20%, a **weighted choice** is a raffle where each option holds that share of the tickets. Any single draw can come out differently, but over many draws the counts settle near the shares. Always taking the largest share instead gives the same answer every time.

```
50 / 30 / 20 → over 100 draws, expect about 50, 30 and 20 (not exactly)
```

THINK OF IT AS

A router that either always takes the top route, or sends traffic in proportion to confidence. The first is repeatable, the second adds variety.

YOUR TURN

The shares are 70 / 20 / 10. In 100 draws, about how many go to the first option?

8 Straight lines and bends

Multiplying and adding only ever produce straight-line behaviour, and a chain of straight steps is still one straight step. A small **bend**, such as “output the input, but never below zero” (written max(0, x)), breaks that, so a chain of steps can follow curves.

```
double, add 3, triple = 3 × (2x + 3) = 6x + 9: still a straight line
```

```
max(0, −2) = 0 and max(0, 3) = 3: a bend
```

THINK OF IT AS

An IF rule: do nothing below a threshold, act above it. Without a step like that, ten calculations collapse into one.

YOUR TURN

Work out max(0, x) for x = −5, 0 and 4.

9 Rescaling

To keep numbers in a comfortable range, subtract their **average** and divide by their **spread** (standard deviation). The result always has average 0 and spread 1, whatever the starting scale.

```
[3, 9, 6]: average 6, spread 2.45 → [−1.22, 1.22, 0]
```

THINK OF IT AS

Putting every sensor on the same scale before comparing them.

YOUR TURN

The numbers are [2, 4, 6]. What is the average? Subtract it from each number.

10 Error and nudging

To find good settings without knowing the answer in advance, **guess, measure the error, nudge, repeat**. Each nudge moves a setting a little in the direction that shrinks the error. This is the idea behind finding the best settings by trial and refinement.

```
prediction = w × 2, target 6. w = 2 → 4 (error 2). w = 2.5 → 5 (error 1). w = 3 → 6 (error 0)
```

THINK OF IT AS

Tuning a rule threshold against past cases: run, check how many it gets wrong, adjust, run again.

YOUR TURN

prediction = w × 3 and the target is 12. If w = 3, what is the error? Which w gives 0?

11 Feedback loops

Take an output and feed it back in as the next input. Each pass uses everything that came before, so small rules can build long results, and the number of passes decides the length of the result.

```
start at 1, double and add 1: 1 → 3 → 7 → 15 → 31
```

THINK OF IT AS

A workflow that calls itself, or interest that earns interest.

YOUR TURN

Start at 2. Double and subtract 1, three times. What do you get?

Three questions to bring

No need to answer them. Just have them in mind while you read.

1. What in your own work could be described as a list of numbers?
2. Where do you already compare two things with a score?
3. Where does a process feed its own output back in as the next input?

Words you will see

Dimension	How many numbers a vector has.
Weight	A number that says how much something counts. A table of weights is a matrix.
Score	One number that rates how good, large or relevant something is.
Probability	A share between 0% and 100%. All the options together add up to 100%.
Function	A rule that turns an input number into an output number.
Iteration	Doing the same step again and again, each time starting from the last result.

Answer key

Try each “Your turn” before you look.

1	[4, 120, 2] (any fixed order works)
2	5 axes, so a 5-dimensional space
3	[3, 4]. It is 5 away; [6, 1] is about 6.1 away
4	$3\times1 + 0\times5 + 2\times4 = 11$
5	$[2\times1 + 1\times4, 2\times3 + 1\times0] = [6, 6]$
6	Equal shares: about 33% each
7	About 70 (not exactly 70)
8	0, 0 and 4
9	Average 4, giving $[-2, 0, 2]$
10	Error 3 (9 vs 12). w = 4 gives 0
11	3, then 5, then 9

Every example uses small, made-up numbers so you can check them by hand.